Assignment-2 Submission Date: 11/11/2021

DEEP LEARNING

(b) Word

2. Explain how a GAN is trained.

There are two aspects that make generative adversarial networks more complex to train than a standard feed forward neural network.

. The generator and the discriminator are really two neural network which must be trained separately, but they also interact so they cannot be trained completely independently of each other.

. It is hard to identify exactly when a generative adversarial network converged

. Since the generator and discriminator have their own separate loss functions, we have to train them separately

. We can do this by alternating between the 2:

- We train the discriminator for one or more epochs, keeping the discriminator weights constant.

We repeat steps 1 and 2 until we determine that the network has converged.

Part 1:- The Discriminator is trained while the Generator is idle. In this phase the network is only forward propagated and no back propagation is done. The Discriminator is trained on real data for n epochs, and see if it can correctly predict them as real. Also, in this phase, the Discriminator is also trained on the fake generated data from the Generator and see if it can correctly predict them as fake.

Part 2:- The Generator is trained while the Discriminator is idle. After the Discriminator is trained by the generated fake data of the Generator, we can get its predicts and use the results for training the Generator and get better from the previous state to try and fool the Discriminator.

The adabove method is repeated for a few epochs and then manually check the fake data if it seems genaine. If it seems acceptable, then the training is stopped. otherwise its allowed to continue for few more epochs.

Ans
14/11/22